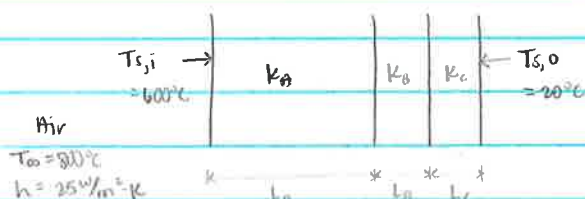


CHE 312 HW #4

3.16, 3.43, 4.28, 4.35

100/100

1) 3.16



$$k_A = 25 \text{ W/m}\cdot\text{K}$$

$$k_C = 60 \text{ W/m}\cdot\text{K}$$

$$L_A = 0.40 \text{ m} \quad L_B = 0.20 \text{ m}$$

$$L_C = 0.20 \text{ m}$$

$$\dot{q} = \frac{T_{s,i} - T_{s,o}}{R_A + R_B + R_C}$$

$$\dot{q} = \frac{T_{\infty} - T_{s,i}}{R_{\text{conv},1}} = \frac{800 - 600}{1/25} = 5000 \text{ W/m}^2$$

$$R_{\text{conv},1} = \frac{1}{h}$$

$$5000 \text{ W/m}^2 = \frac{T_{s,i} - T_{s,o}}{R_A + R_B + R_C} = \frac{600^\circ\text{C} - 20^\circ\text{C}}{\frac{0.40}{25 \text{ W/m}\cdot\text{K}} + \frac{0.20}{k_B} + \frac{0.20}{60 \text{ W/m}\cdot\text{K}}}$$

$$R_A = \frac{L_A}{k_A}$$

$$5000 \text{ W/m}^2 = \frac{580^\circ\text{C}}{0.16 \text{ W/m}\cdot\text{K} + \frac{0.20}{k_B} + 0.0033 \text{ W/m}\cdot\text{K}}$$

$$k_B = 2.069 \text{ W/m}\cdot\text{K}$$

2) 3.43

$$a. \dot{E}_g = I^2 R$$

$$= (20 \text{ A})^2 (0.01 \frac{\Omega}{\text{m}}) = 4.0 \frac{\text{W}}{\text{m}} = 4.0 \text{ W/m} \quad r = 0.001 \text{ m}$$

$$\dot{q} = \frac{\dot{E}_g}{\text{area}} = \frac{4.0 \frac{\text{W}}{\text{m}}}{\pi (0.001 \text{ m})^2} = 1.27 \times 10^6 \text{ W/m}^2$$

$$b. \dot{E}_g = \dot{q}' = \dot{q}'_{\text{conv}} + \dot{q}'_{\text{rad}} = \pi D h (T - T_{\infty}) + \pi D \epsilon \sigma (T^4 - T_{\text{sur}}^4)$$

$$T_{\infty} = T_{\text{sur}} = 20^\circ\text{C} = 293 \text{ K}$$

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4$$

$$C = 1.25 \text{ W/m}^{3/4}\cdot\text{K}^{0.4}$$

$$\epsilon = 0.3$$

$$h = C [(T - T_{\infty})/D]^{1/4}$$

$$4.0 \text{ W/m} = \pi (0.002) 1.25 \text{ W/m}^{3/4}\cdot\text{K}^{0.4} \left(\frac{T - 293}{0.002} \right)^{1/4} (T - 293) + \pi (0.002) (0.3) (5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) (T^4 - 293^4)$$

$$\text{Using solver in calculator: } T = 331.0 \text{ K}$$

$$c. \dot{E}_g = \pi D_o h (T_{s,o} - T_{\infty}) + \pi D_o \sigma (T_{s,o}^4 - T_{\text{sur}}^4) \epsilon$$

$$4.0 \text{ W/m} = \pi (0.006) (1.25) \left(\frac{T_{s,o} - 293}{0.006} \right)^{1/4} (T_{s,o} - 293)$$

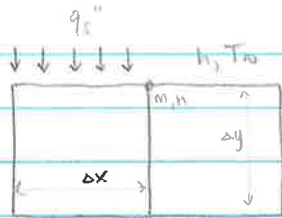
$$+ \pi (0.006) (5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) (T_{s,o}^4 - 293^4) (0.9)$$

* same as part B, using solver in calculator

$$T_{s,o} = 307.79 \text{ K}$$

4

4.35



$$q_1 + q_2 + q_3 + q_4 + q_5 = 0$$

$$\Delta x = \Delta y$$

$$q_1 = q_s'' \cdot A = q_s'' \frac{\Delta x}{2} \cdot 1$$

$$\text{heat convection: } q_2 = h(T_\infty - T_{m,n}) \frac{\Delta x}{2} \cdot 1$$

$$\text{heat conduction: } q_3 = \frac{k(T_{m-1,n} - T_{m,n})}{\Delta x} \cdot \frac{\Delta y}{2} \cdot 1 \Rightarrow \frac{k(T_{m-1,n} - T_{m,n})}{2} \quad \checkmark$$

$$q_4 = \frac{k(T_{m+1,n} - T_{m,n})}{\Delta x} \cdot \frac{\Delta y}{2} \cdot 1 \Rightarrow \frac{k(T_{m+1,n} - T_{m,n})}{2}$$

$$q_5 = \frac{k(T_{m,n-1} - T_{m,n})}{\Delta x} \cdot \Delta y \cdot 1 \Rightarrow k(T_{m,n-1} - T_{m,n})$$

$$\Rightarrow q_s'' \frac{\Delta x}{2} + h(T_\infty - T_{m,n}) \frac{\Delta x}{2} + \frac{k}{2}(T_{m-1,n} - T_{m,n}) + \frac{k}{2}(T_{m+1,n} - T_{m,n}) + k(T_{m,n-1} - T_{m,n}) = 0 \quad \checkmark$$

(25/25)